



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

SEMESTER 1/20252026

ARTIFICIAL INTELLIGENCE (SECJ3553)

SECTION 03

GROUP 2

**PROJECT:
ECOROUTE**

**THEME:
SMART CITY**

**LECTURER:
DR NORSHAM IDRIS**

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Problem Statement

The current waste management framework in Malaysian cities still follows a rigid schedule for example collections on every Monday and Thursday without considering how much rubbish is being produced at any given time. This inefficiency forces the City Councils often end up spending unnecessary fuel and budget to collect bins that are barely filled, while other high-density areas experience overflowing bins during peak periods. This not only wastes resources but also leads to unpleasant and unhygienic conditions that can affect public health and the well-being of residents.

AI Solution

EcoRoute is an AI-powered system designed to transform traditional waste collection into a fully intelligent, demand-driven process. Instead of following fixed schedules, EcoRoute uses real-time sensor data and predictive analytics to determine when each rubbish bin actually needs servicing. By continuously monitoring fill-levels, the system identifies bins that are almost full and prioritizes them for collection.

A route optimisation engine then generates the most efficient collection paths for waste trucks, reducing unnecessary trips to empty bins and preventing overflowing bins in high-usage areas. The AI model also learns from historical waste patterns, such as daily trends, seasonal surges, or location-specific usage to predict when individual bins are likely to reach 80% capacity, enabling proactive planning rather than reactive cleanup.

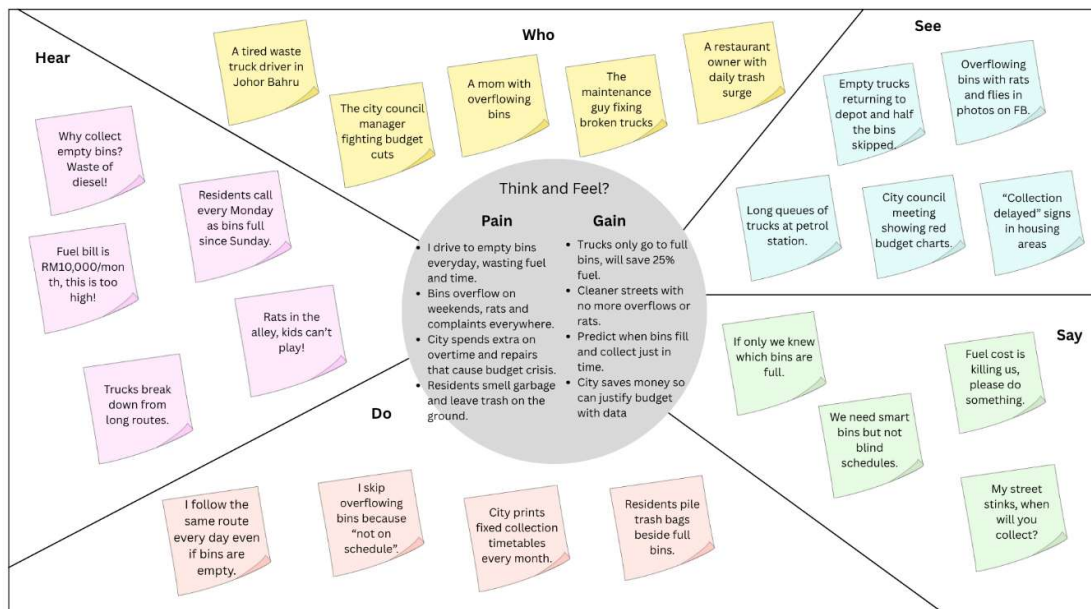
Ultimately, EcoRoute helps local councils and city management teams reduce operational and fuel costs, supports waste collection crews with clearer and more practical routes, and ensures cleaner, more hygienic neighbourhoods for residents. The solution contributes directly to sustainability goals by decreasing carbon emissions and minimizing wasted resources, making waste management smarter, greener, and more efficient.

Goal of AI Solution

The main goal of *EcoRoute* is to enhance the efficiency and sustainability of city waste management in Malaysian cities by using AI to monitor and predict bin fill-levels in real time. Through this system, the project aims to:

1. Support waste management crews and city administrators in identifying which bins need collection, reducing unnecessary trips.
2. Improve operational efficiency by generating AI-optimized collection routes based on real-time and estimated bin occupancy.
3. Minimise fuel consumption and carbon emissions, contributing to a greener and more sustainable urban environment.
4. Empower city authorities with data-driven insights to optimize waste collection planning, resource allocation, and overall service efficiency.

Process of Emphasis in Design Thinking



We empathized with waste crews, city councils and residents by stating their daily struggles in the sticky note sketch. The 'Pain and Gain' circle shows pains like wasted fuel and overflowing bins, with gains like smart routes and cleaner streets. 'Who' includes drivers and residents, 'Hears' captures complaints, 'Does' notes fixed schedules, 'Sees' spots empty trucks, and 'Says' reflects frustration like 'Why collect empty bins?' This aims for a predictive, on-demand collection system.

Process of Define in Design Thinking

User	Need	Insight
City Council	Need 1: To optimize the waste collection budget by reducing fleet operational costs like fuel, vehicle maintenance, and staff overtime Need 2: Needs verifiable data on collection efficiency to report to the city council and justify her department's budget.	Insight 1: The current fixed-route system is fundamentally inefficient because it treats all bins as equal. This fix scheduling wastes significant resources by forcing crews to drive to stop at, and service bins that are only 20% full, while simultaneously failing to cope with sudden surges in other areas.
A resident living in a high-density apartment complex	Need 1: To have reliable and sanitary access to waste disposal bins that are not overflowing. A resident needs to be able to dispose of his trash without experiencing nasty smells, rats or having to leave his trash bags on the ground, which he feels is a public health hazard.	Insight 1: The problem is not only that the bins get full, but that they get predictably. The existing static schedule such as collection every Monday or Thursday totally ignores the residence's actual usage habits such as higher waste generation on weekends. The bins are likely to overflow due to the inflexible schedules.
Waste Truck Driver	Need 1: To have shorter, smarter daily routes that reduce physical fatigue and driving time. Need 2: To avoid unnecessary maintenance on trucks from excessive, low-value driving, which leads to breakdowns.	Insight 1: The inefficiency of the fixed-route system has a direct human cost. Drivers are wasting an average of 2 hours per day driving to and stopping at empty bins, leading to burnout and fatigue. Insights 2: Financial effect occurs. On days when high-density bins overflow due to the bad schedule, workers

		are forced into overtime to clean up the rubbish, which directly increase the City Council's operational costs.
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ASSIGNMENT 1:
EcoRoute

THEME:
SMART CITY

LECTURER:
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Knowledge Representation

1. **High-Priority Bin Collection:** Keep monitoring the bin until it exceeds a certain level of amount.

*IF (bin_fill_level("Bin_123") > 85%) AND (bin_status("Bin_123") != "Damaged")
THEN add_to_route("Bin_123", "HighPriority")*

Explanation:

This KR is solving the City Council's primary need for operational efficiency. EcoRoute uses its sensors to continuously check each bin's fill level. According to this rule, a bin should only be taken into consideration for a collection route if its fill level exceeds a high threshold of 85%. The outdated, ineffective method of maintaining empty or barely used bins is immediately stopped by this logic. The AI can determine the most effective way to service only these high-value targets by generating "HighPriority" tasks exclusively for full bins. Another level of intelligence is added by the AND condition (! = "Damaged"), which stops the system from wasting a truck's time by sending it to an unusable location. This prevents unsuccessful collection attempts, which are a significant reason for fuel and workers wasted.

Predicate Logic/ First Order Logic:

$$\forall x((FillLevel(x) > 0.85) \wedge \neg bin_status(x, "Damaged")) \\ \rightarrow AddRoute(x, HighPriority)$$

For any bin (x), if the fill level of bin (b) is greater than 85, AND the status of bin (b) is NOT "Damaged", THEN the system will add bin (b) to the collection route with "HighPriority".

- 2. Predictive Overflow Prevention:** Proactively monitoring bins during peak periods like weekends and public holidays.

*IF (current_day = "Weekend" OR "PublicHoliday") AND
(historical_overflow_risk("Bin_123", current_day) = "High") AND
(bin_fill_level("Bin_123") > 60%) THEN add_to_route("Bin_123",
"MediumPriority")*

This KR focuses on preventing bin overflow by leveraging predictive insights and contextual factors such as weekends and public holidays. The technology analyzes past data and current sensor readings to identify bins that are likely to overflow during high-demand days. According to this rule, if the current day is a weekend or public holiday, and the historical overflow risk for a bin like “Bin_123” is recorded as “High,” and its current fill level exceeds 60%, then the system will automatically add the bin to the collection route with “MediumPriority.” By ensuring that trash cans in crowded places are serviced before they overflow, this preventive thinking improves cleanliness and reduces citizen complaints.

Explanation:

This rule directly solves the resident's need. Instead of waiting for the bin to hit 85%, this rule uses three conditions:

1. Calendar: It checks if today is a "Weekend" or "PublicHoliday".
2. Historical Data: It checks its memory to see if this specific bin has a "High" risk of overflowing on these days.
3. Lower Threshold: It uses a lower fill level (> 60%) to act before the bin becomes critical.

Predicate Logic/ First Order Logic (FOL):

$$\begin{aligned} \forall x \forall d \text{ (OnDay}(d) \wedge (\text{Weekend}(d) \vee \text{PublicHoliday}(d)) \\ \wedge \text{HistoricalOverflowRisk}(x, \text{"High"}) \wedge \text{FillLevel}(x) > 0.60) \\ \rightarrow \text{AddRoute}(x, \text{"MediumPriority"}) \end{aligned}$$

For any day (d) and any bin (x), if the day (d) is a Weekend OR a Public Holiday, AND the historical overflow risk for that bin (x) is "High", AND the fill level of bin (x) is greater than 0.6 (60%), THEN the system will add bin (x) to the collection route with "MediumPriority". This enables predictive scheduling that helps the waste management system handle peak times efficiently and avoid overflow incidents before they occur.

- 3. Commercial Zone Priority:** If a bin is in a commercial zone and fill level >70%, assign it to the nearest available truck.

*IF Location("Bin_123", "Commercial") AND FillLevel("Bin_123") > 0.70
THEN AssignTruck("Bin_123", NearestTruck)*

Explanation:

EcoRoute uses GPS to classify all bins by zone type which are residential, commercial or industry. Commercial areas like City Square or restaurants generate significantly more waste daily than homes. This rule triggers when a commercial bin fill level hits 70%, a threshold lower than residential to account for faster filling. For example, Bin_789 near a food court at 75% full, then system automatically locates the nearest available truck. This minimizes response time, prevents overflow in high-traffic public areas and balances truck workload. It directly supports the city council's goal of efficient fleet management.

Predicate Logic / First Order Logic (FOL):

$$\forall x (Location(x, "Commercial") \wedge FillLevel(x) > 0.70) \\ \rightarrow AssignTruck(x, NearestAvailable)$$

For any bin (x), if it is located at a commercial zone AND its fill level exceeds 70%, then assign it to the nearest available truck. The $\forall x$ applies to all commercial bins. The $\wedge$ ensures both location and fill conditions. The $\rightarrow$ triggers dynamic truck assignment. This support real-time efficiency and driver workload balance.

- 4. Driver Route Notification:** If a route is generated and the truck is idle, send the route to the driver's tablet.

*IF RouteGenerated(Truck_A) AND TruckStatus(Truck_A, "Idle")
THEN SendRoute(Truck_A, DriverApp)*

Explanation:

EcoRoute generates optimized routes only when needed and sends them instantly via mobile app to drivers. This rule will wait for the truck to complete its current task and enter "Idle" status before pushing the new route. Then, driver receives a clear, interactive map: "Bin_123 to Bin_456 to Depot (45 min, 12 km)" with turn-by-turn GPS. By doing so, no more paper schedules, radio calls, or confusion. This improves driver satisfaction, reduces downtime and ensures real-time execution of dynamic plans. It supports the driver's need for shorter and smarter daily routes.

Predicate Logic / First Order Logic (FOL):

$$\forall t (RouteGenerated(t) \wedge TruckStatus(t, "Idle")) \\ \rightarrow SendRoute(t, DriverApp)$$

For any truck (t), if a route has been generated for (t) AND (t) is currently idle, then send the route to the driver's app. The $\forall t$ applies to all 20 trucks in the fleet. The $\wedge$ ensures timing is correct. The $\rightarrow$ triggers push notification. This completes the on-demand collection loop.

- 5. Truck Overloading Prevention:** Detect current truck capacity to avoid from assigning it to another bin when maximum capacity is nearly reached.

IF (TruckLoad ("Truck_A") > 90% THEN ReassignRemainingBins ("Truck_A", NearestAvailableTruck)

Explanation:

EcoRoute uses a weight sensor to continuously check current load percentage of the waste collection truck. This rule is used as an indicator to reassign the bin to be collected, when the current assigned truck's load surpasses the threshold of 90%. It will find the nearest available truck and transfer all remaining bins of Truck_A as Truck_A will immediately stop collecting bins. The system ensures a safe waste transportation and increases efficiency by preventing minimizing unexpected delays or hazard, such as mechanical strain and braking problems due to overweight truck.

Predicate Logic/ First Order Logic (FOL):

$\forall t (TruckLoad(t) > 0.90) \rightarrow ReassignRoute(t, NearestAvailableTruck)$

For any truck (t), IF the truckload is over 0.90, THEN the system will reassign the bin route from truck (t) to the nearest truck which is available. This ensures a safe operating limit and prevent overloading incidents.

6. **Route Optimization Trigger:** If more than 5 high-priority bins are located in the same zone, automatically generate an optimized collection route using the Traveling Salesperson Problem (TSP) algorithm.

*IF CountHighPriorityBins (Zone_A) > 5
THEN GenerateRoute (Truck_X, TSP_Algorithm)*

Explanation:

EcoRoute monitors bin clusters every hour using GPS coordinates. When 6 or more high-priority bins appear in one zone, for example, Taman Ungku Tun Aminah night market area, the fixed-route system becomes highly inefficient. This rule triggers the TSP algorithm to calculate the shortest possible path that visits all full bins and returns to the depot. For example, 8 bins in Zone_A that used to take 21 km under fixed schedule are reduced to only 13 km. This single rule can save 2 to 3 hours and 30 to 40% fuel per truck per day, directly contributing to the city council's budget reduction goal.

Predicate Logic/ First Order Logic (FOL):

$$\forall z \forall t (Zone(z) \wedge CountHighPriority(z) > 5 \wedge AvailableTruck(t)) \\ \rightarrow GenerateRoute(t, z, TSP)$$

For any zone (z) and truck (t), if the zone contains more than 5 high-priority bins AND truck (t) is available, then generate an optimized route using TSP. The quantifiers $\forall z \forall t$ cover all zones and trucks and $\wedge$ combines the conditions. The $\rightarrow$ activates the optimization engine. This rule transforms scattered collection points into the shortest possible path.

Knowledge Representation Involved to Achieve Goals

EcoRoute uses a declarative, hybrid knowledge representation strategy focused on providing actionable, real-time waste management decisions. This is achieved primarily through Logical Representation, specifically First-Order Logic, FOL and Production Rules. This combination forms the core of EcoRoute, enabling it to do more than just monitor bins and instead make smart decisions for waste collection routes.

Logical Representation (FOL)

EcoRoute employs First-Order Logic to formally describe relationships, constraints, and conditions between bins, trucks, zones, and times. For example, it can express rules like:

$$\forall x((\text{FillLevel}(x) > 0.85) \wedge \neg \text{bin_status}(x, \text{"Damaged"})) \\ \rightarrow \text{AddRoute}(x, \text{HighPriority}))$$

This ensures that the system is logically consistent, clearly defines its domain, and allows precise reasoning about operational decisions. It provides a solid foundation to model complex relationships, such as predicting overflow, prioritising bins in commercial zones, and assigning trucks optimally.

Production Rules

Production Rules serve as the operational mechanism to execute actions based on conditions. These are implemented as IF–THEN rules, defining thresholds and triggers for tasks such as High-Priority Bin Collection, Predictive Overflow Prevention, and Driver Route Notification.

For example:

*IF Location("Bin_123", "Commercial") AND FillLevel("Bin_123") > 0.70
THEN AssignTruck("Bin_123", NearestTruck)*

This approach is modular, interpretable, and easy to maintain, allowing new rules to be added as city requirements change. Production Rules, combined with FOL, enable EcoRoute to act intelligently in real time, dynamically optimizing collection routes and reducing fuel consumption, operational costs, and overflow incidents.

Assignment 1 Peer Review Form

Evaluator: Choh Jing Yi

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Brendan Chia Yan Fei	4	
Anis Safiyya Binti Janai	4	
Lim Yu Han	4	

Evaluator: Brendan Chia Yan Fei

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Choh Jing Yi	4	
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Evaluator: Anis Safiyya Binti Janai

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SECTION 03

GROUP 2

ASSIGNMENT 2:

EcoRoute

THEME:

SMART CITY

LECTURER:

DR NORSHAM IDRIS

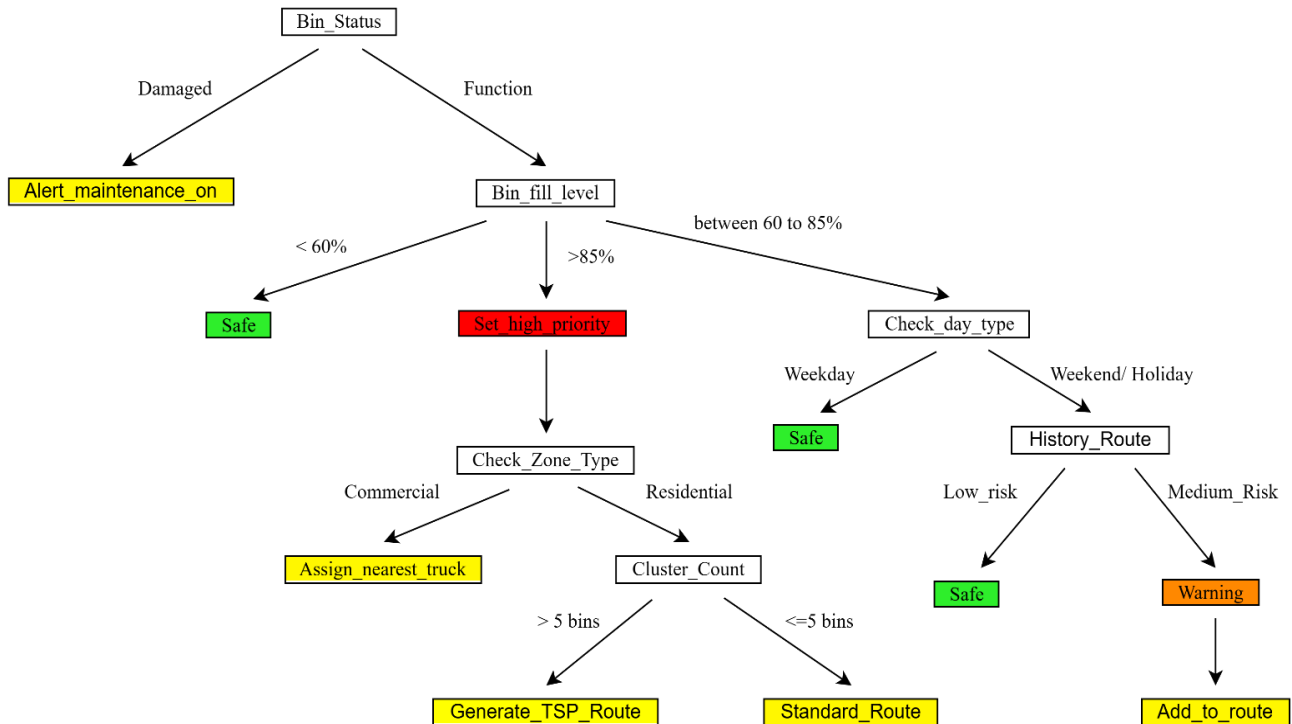
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1.0 Problem Formulation by State Space Graph

1.1 Bin Management



1.2 Explanation States and Actions

As proposed in our project proposal document and Assignment 1, EcoRoute is a waste management system based on AI that dynamically observes the condition of the bins, estimates the possibility of bin overflow, and provides the best routes to be followed by the trucks. The state space of the EcoRoute system can be described through a number of critical states. The transition from the states can be described through actions. The combination of states and actions guarantees efficient waste collection.

States	Bin_Status - Damaged: The bin cannot be serviced and requires maintenance before collection. This prevents wasted trips and ensures resources are not allocated to unusable bins. - Function: The bin is operational and can be monitored for fill levels, enabling the system to make real-time decisions. Bin_Fill_Level < 60%: Safe, no immediate collection required.
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	• 60 – 85%: Medium risk, monitored closely to prevent overflow. • > 85%: High priority, required immediate collection. This state directly implements the KR1 rule from Assignment 1 which is $\forall x((FillLevel(x) > 0.85) \wedge \neg bin_status(x, "Damaged")) \rightarrow AddRoute(x, HighPriority)$ 3. Day_Type • Weekday: Normal collection rules apply. • Weekend / Holiday: Higher risk of overflow due to increased waste generation. Predictive scheduling is triggered, lowering thresholds to act earlier. 4. Zone_Type • Commercial: Bins fill faster, lower threshold (70%) applied. • Residential: Bins fill slower, higher threshold (85%) applied. This ensures fairness and efficiency across different urban zones. 5. History_Route • Low Risk: Bin rarely overflows, standard scheduling applies. • Medium / High Risk: Bin frequently overflows, predictive scheduling required. This state uses historical data to anticipate problems before they occur. 6. Cluster_Count • ≤ 5 bins: Standard route assignment. • > 5 bins: Route optimization using the Traveling Salesperson Problem (TSP) algorithm. This ensures trucks take the shortest possible path when servicing multiple bins.
Action	1. Alert_Maintenance_On: Triggered when a bin is damaged. The system alerts maintenance teams to repair the bin before collection, preventing wasted fuel and time. 2. Set_High_Priority: Triggered when a bin's fill level exceeds 85%. The bin is marked for immediate collection, ensuring hygiene and preventing overflow. 3. Add_To_Route: Adds bins that meet threshold conditions to the truck's collection route, ensuring only necessary trips are made. 4. Assign_Nearest_Truck: Allocates the closest available truck to a bin, reducing travel time, fuel consumption and driver fatigue. 5. Generate_TSP_Route: When more than 5 high-priority bins are detected in a zone, the system applies the TSP algorithm to generate the most efficient route. This reduces distance traveled and operational costs. 6. Standard_Route: Used when fewer than 5 bins require collection, following normal scheduling.

1.3 Problem Formulation and Explanation

Critical	Details	Explanation
Initial state	Bin_Status = Function Bin_Fill_Level < 60% Day_Type = Weekday Zone_Type = Residential	The system starts with a bin that is operational, has a low fill level, and is in a standard monitoring state
Action	Alert_Maintenance_On Set_High_Priority Add_To_Route Assign_Nearest_Truck Generate_TSP_Route Standard_Route	These are the specific operations the system can perform to transition between states, such as flagging a bin for repair, changing its collection priority, or optimizing the truck's path.
Goal	Bin_Fill_Level < 60% Bin_Status = Function	The desired outcome is that all bins are emptied and remain functional, preventing overflow and ensuring system efficiency.
Path Cost	Distance traveled by trucks Fuel consumption Time consumption	Longer routes increase fuel usage and time consumption. Higher fuel usage increase overall operational cost. Longer time to complete bin collection lead to inefficiencies and delays.

Solution: sequence of actions leading from initial state to goal state

1. Monitor Bin_fill_level

- Ensure the bin is functional, and the fill level increases over time.

2. Set_high_priority

- When Bin_fill_level exceeds 85%, the system marked the bin as high priority.

3. Assign_nearest_truck

- The system assigns the bin to the nearest truck that is available to reduce travel costs.

4. Generate_TSP_Route

- Use TSP optimization when the high-priority bin in the zone is more than 5.

5. Truck Collects Waste

- Truck follows the route generated and empties the bin.

6. Update Bin Status

- After collection, Bin_Fill_Level is reset to < 60% and Bin_Status remains function.

Assignment 2 Peer Review Form

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1.0 PEAS Model

Performance measure	1. Efficiency: Minimize total fuel consumption (L) and distance travelled (km). 2. Speed: Reduced time for collection cycles. 3. Reliability: Zero overflow incidents (preventing the "High Risk" state).
Environment	1. Roads: Urban traffic network. 2. Zones: Residential & Commercial areas. 3. Temporal: Weekdays vs. Weekends. 4. Online Platform: Cloud database & Driver App.
Actuators/Effectors	1. Display: Driver navigation interface. 2. Virtual: Route Optimization Engine (TSP). 3. Alerts: Push notifications to drivers. 4. Database: Status updates such as Set High Priority.
Sensors	1. Ultrasonic: Bin fill-level sensors. 2. GPS: Truck and Bin location trackers. 3. Weight: Truck load sensors.

P - Performance Measure

- Efficiency: The primary measure is minimizing the Path Cost. The agent is successful if it reduces the litre of diesel used per kilogram of waste collected.
- Speed: The agent aims to minimize the total duration of a collection round, ensuring drivers complete their shifts without unnecessary overtime.
- Accuracy: The agent judges its own accuracy by its ability to predict and collect waste before a bin overflows. If a bin reaches 100% capacity before a truck arrives, the agent has failed.

E - Environment

- **Roads:** The agent must navigate the actual road network of the city, which includes traffic lights, congestion, and varying distances between bins.
- **Zones:** The environment is divided into specific zones such as residential or commercial zones which dictate how fast waste accumulates.
- **Temporal:** The environment changes based on time. A weekend or public holiday environment has more waste generation than weekday.
- **Online Platform:** The agent also presents on the cloud server where it processes data from the peer-to-peer network of bins and trucks.

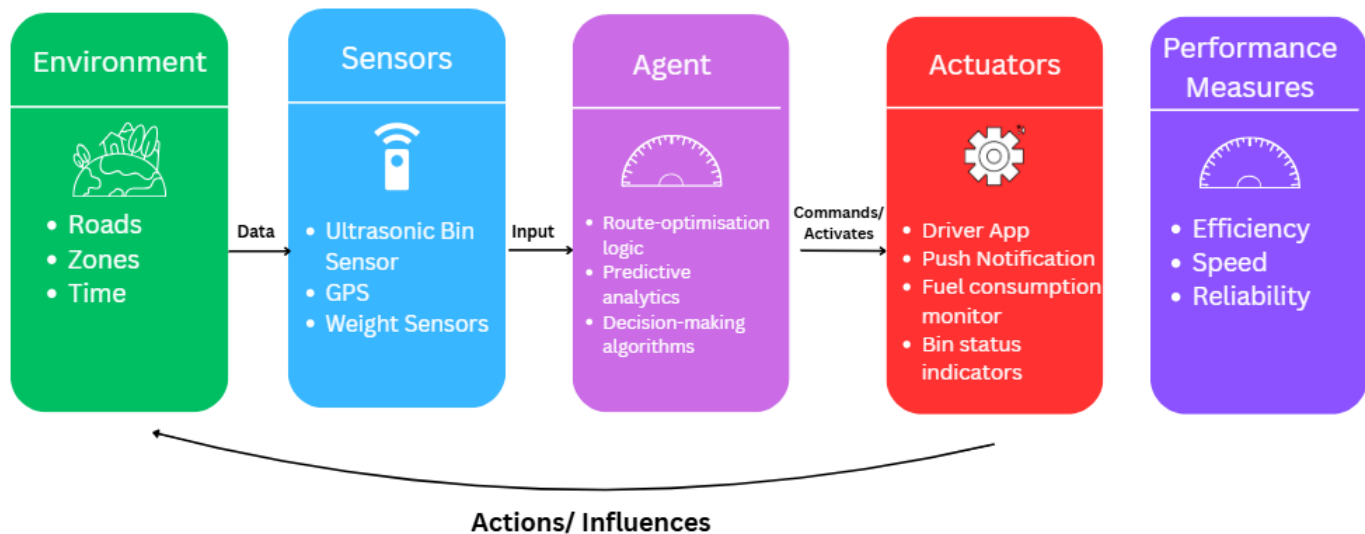
A - Actuators

- **Display:** This agent represents driver interface which displays the optimized path and turn-by-turn navigation instructions.
- **Virtual Mechanisms:** The TSP Algorithm acts as an internal actuator. It physically reorders the sequence of stops in the database for more efficient route.
- **Alerts:** The agent triggers Push Notifications when high priority bin detected. This mechanism notifies the driver to collect high priority bins.
- **Database:** The agent acts by changing the state of a bin in the system from "Monitor" to "High Priority".

S - Sensors

- **Ultrasonic Sensors:** These are the primary "eyes" inside the bin lids. They measure the distance to the waste pile to determine the Bin Fill Level.
- **GPS:** The agent uses GPS modules to perceive the coordinates of every truck and bin. This allows it to calculate proximity for the "Assign Nearest Truck" rule.
- **Weight/Load Sensors:** Sensors on the truck's suspension perceive the current weight of the vehicle, allowing the agent to detect if the truck is full.

2.0 Relations Between Each Property



The EcoRoute agent operates through a continuous cycle that integrates the PEAS parameters to achieve its optimisation objectives. The process begins in the Environment, which provides raw data perceived by sensors, such as Ultrasonic Bin Sensors, GPS, and Weight Sensors. These sensors provide real-time input to the Agent, which serves as the system's brain. Within the Agent block, the system applies Route-optimization logic, predictive analytics, and decision-making algorithms to process the incoming data.

Once a decision is reached, the Agent sends commands to activate the Actuators, which include the Driver App, Push Notifications, Fuel Consumption Monitors, and Bin Status Indicators. These actuators execute the necessary actions that directly influence the Environment. For example, by guiding a truck to a high-priority zone for collection. Finally, the entire operation is evaluated against Performance Measures, such as Efficiency, Speed, and Reliability, to ensure the results meet the system's goals. This closed feedback loop allows the agent to constantly refine its logic and decision rules, ensuring continuous improvement in smart waste management.

3.0 Representation of PEAS Properties in the Proof of Concept

In the Proof of Concept (POC) for EcoRoute, the physical components of the smart waste management system are simulated through software modules and mock data. This allows the demonstration of the AI agent's intelligent behavior without requiring physical trucks or installed sensors. The following subsections detail how each element of the PEAS model is represented within the software prototype.

3.1 Performance Measure Representation

Efficiency in the POC is measured by minimizing path cost, represented through total distance travelled and estimated fuel consumption. Distance is calculated using GPS-based route tracking, while fuel usage is estimated using a fixed fuel-per-kilometre rate.

Performance is evaluated by comparing fuel consumption per kilogram of waste collected between baseline and optimized routes.

Speed is represented by the total duration of each collection cycle. The POC records start and end times of routes and incorporates simulated traffic conditions to reflect peak and off-peak periods. The agent is considered effective if it reduces total collection time while completing routes within driver shift limits.

Reliability is measured by the system's ability to prevent bin overflow. Bin fill levels are continuously monitored, and overflow incidents are recorded if bins reach full capacity before collection. The POC demonstrates reliability when high-risk bins are collected in time, resulting in zero or reduced overflow events.

3.2 Environment Representation

The urban road network is modeled as a graph with distances and time-based traffic adjustments. Residential and commercial zones are defined with different waste generation rates, influencing bin fill behavior. The variation between weekdays and weekends affects both waste accumulation and traffic patterns.

The online platform environment is hosted on a cloud platform that integrates a central database, a route optimisation service and a driver application.

3.3 Actuator Representation

The driver interface acts as the primary actuator by displaying optimised routes. A route optimization engine functions as a virtual actuator by dynamically reordering collection stops using a TPS-based approach. Push notifications are triggered when high-priority bins are detected, alerting drivers in real time. The agent also updates bin states in the database, reflecting actions such as prioritization and collection completion.

3.4 Sensor Representation

Ultrasonic sensors are simulated to provide real-time bin fill-level readings based on zone-specific waste growth. GPS sensors track truck and bin locations, enabling distance calculation and nearest-truck assignment. Truck load sensors simulate weight changes during collection, allowing the agent to monitor capacity and adjust routes accordingly.

4.0 Behaviors of Agent To Achieve The System Goals

As an intelligent agent, the EcoRoute system always follows the cycle of perception, reasoning and action to achieve the goals of reducing operating costs, improving productivity and avoiding bin overflow. With its goal-oriented and adaptable behavior, this intelligent body can respond quickly to environmental changes.

4.1 Perception Behavior

First of all, the intelligent body uses the information obtained by the sensor to perceive the surrounding environment. The GPS sensor provides the location information of the bins and garbage collection trucks, while the ultrasonic sensor installed in the bin measures the filling percentage of the garbage bin in real time. In addition, the load sensors of the garbage truck will track the remaining capacity of the vehicle. Then, the intelligent agent can update the current status record of the system according to the sensor data.

4.2 Reasoning and Decision-Making Behavior

The agent uses the perceived data to classify the status of the bin according to the predefined knowledge representation rules. For example, when the bin's fill level exceeds the preset threshold, the agent will mark it as a high-priority bin.

After that, state space search algorithm is used to analyze the possible action sequence, and selects the sequence with the lowest path cost while considering factors such as truck availability, fuel consumption, urgency and driving distance.

The decision-making process will also be affected by environmental factors. Due to the high amount of garbage generated on weekends and holidays, the agent reduces the collection threshold. Similarly, due to the faster accumulation of garbage in commercial areas, the collection priority of garbage bins is higher than the garbage bins in residential areas.

4.3 Action Execution Behavior

After making a decision, the intelligent body uses the actuator to perform a series of actions. These actions include:

- Run the route optimization engine and rearrange the collection priority
- Assign high-priority garbage bins to the nearest available trucks
- Sending notifications and route updates to drivers
- Updating bin status in the database after collection or assigning priorities

These actions will have a direct impact on the environment and push the system to achieve the expected goal state.

4.4 Goal Achievement and Feedback Loop

After the waste collection is completed, the agent update the environment by recording the collection results to the database and resetting the bin fill level. The agent then integrates these updated data into its subsequent perception loop.

The agent maintains a feedback loop by continuously tracking the results and adjusting the system status as needed, thus, enhancing the future decision-making ability. This behavior ensures the continuous efficiency and reliability of the system while preventing the overflow of the bin.